

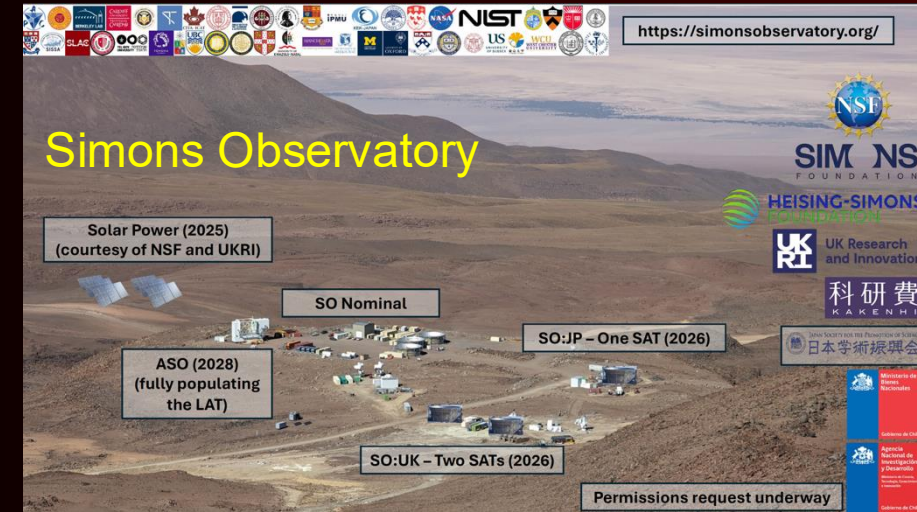
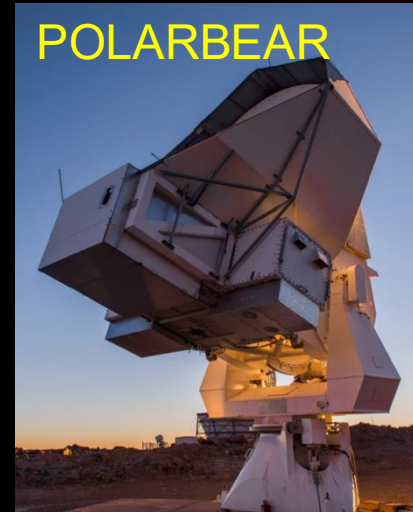
Qubits for light dark matter searches

Masashi Hazumi

National Central University
&
INFN Ferrara Section

My collaboration with Andrew

- 31 co-authored publications (POLARBEAR, Simons Observatory)
- Hosted Andrew as a visiting scientist at QUP, KEK, in Autumn 2024



Who Am I?

**KEK Sabbatical visit
Autumn 2024**



Celebrating Andrew at 60 in the traditional Japanese way ("Kanreki")



AI image
(by Taeko Hazumi)

Qubits for light dark matter searches

Masashi Hazumi

National Central University
&
INFN Ferrara Section

References

Light dark matter search with NV centers in diamond

1. "Light dark matter search with nitrogen-vacancy centers in diamonds," S. Chigusa, MH, E. D. Herbschleb, N. Mizuochi, K. Nakayama, JHEP 03 (2025) 083
2. "Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection," S. Chigusa, MH, E. D. Herbschleb, Y. Matsuzaki, N. Mizuochi, K. Nakayama, Phys.Rev.D 111 (2025) 7, 075028
3. "Hybrid-spin decoupling for noise-resilient DC quantum sensing," S. Chigusa, MH, E. D. Herbschleb, Y. Matsuzaki, N. Mizuochi, K. Nakayama, arXiv:2511.16732

Collaborators



So Chigusa
(MIT)

Particle
Physics



Kazunori
Nakayama
(Tohoku U.)

Particle
Physics



Yuichiro
Matsuzaki
(Chuo U.)

Quantum
Information
Science



David
Herbschleb
(Kyoto U. & KEK)

Quantum
Materials
Research



Norikazu
Mizuochi
(Kyoto U.)

Quantum
Materials
Research

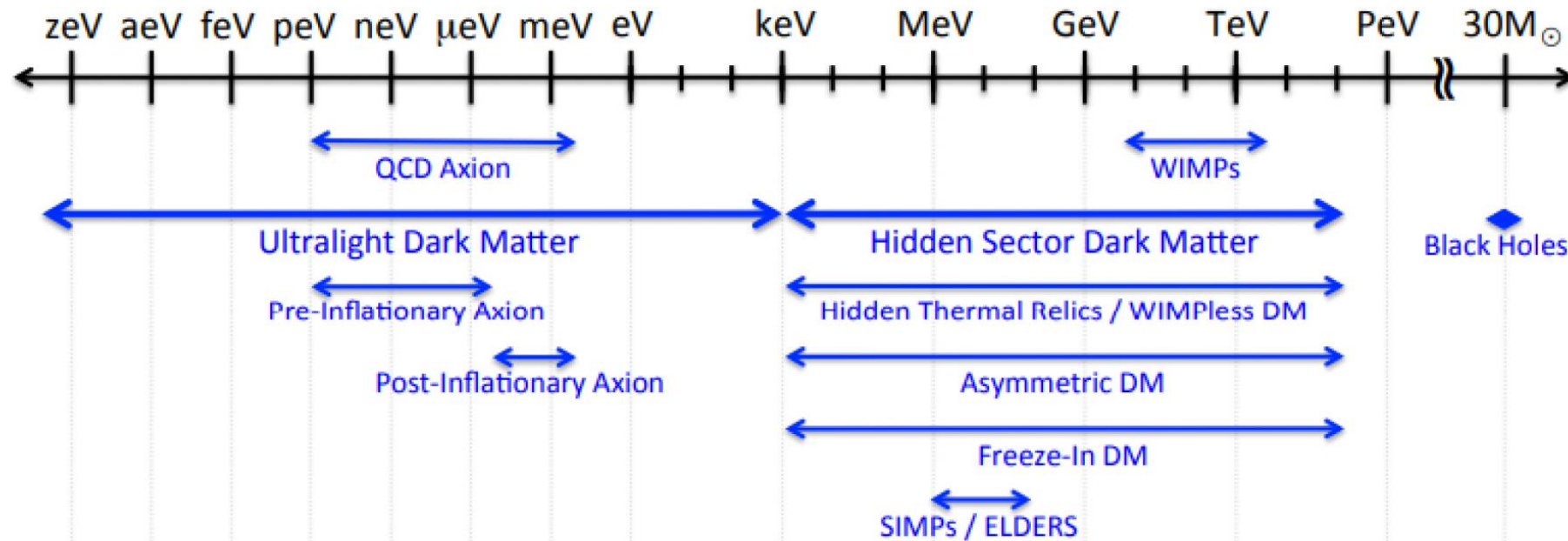
Maximize Opportunities for Discovery: **Delve Deep, Search Wide** (Snowmass 2021 Conclusion)

For all masses, all couplings

huge discovery space in dark matter

arXiv:2210.01770

Mass of DM



US Cosmic Visions: New Ideas in Dark Matter 2017 : Community Report

What could be the next game changer?

“Quantum 2.0”

- ① Quantum computing with **entangled qubits**, the main driver
- ② Quantum communications, or “quantum internet”
- ③ Quantum sensing
- ④ Quantum materials

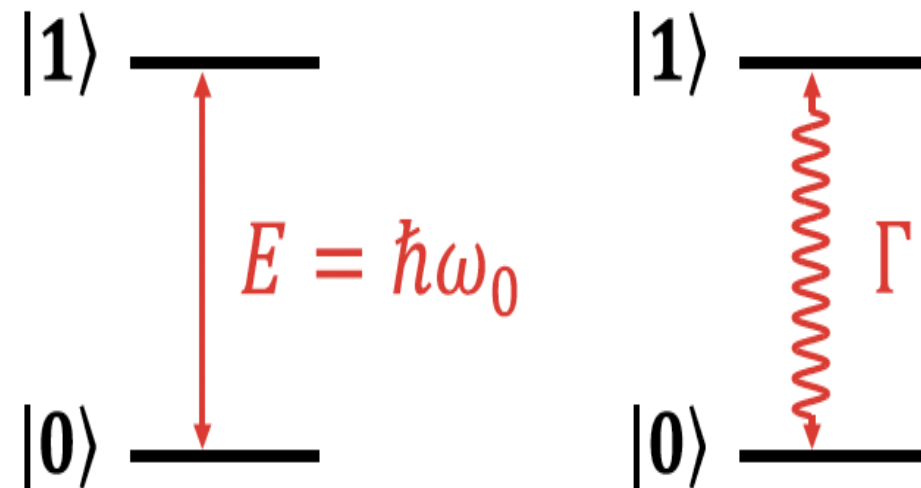
QURECA, <https://www.quireca.com/quantum-initiatives-worldwide/>.

Global investment in quantum technologies continues to grow with current worldwide investments exceeding **\$40 billion**. Overall, the global quantum technology market is projected to reach **\$106 billion by 2040**.

Qubit as quantum sensor

Requirements

1. Discrete quantum states
2. Initialization and readout capability
3. Can be coherently manipulated, typically by time-dependent fields.
4. Can interact with a physical quantity of interest, leading to a change of E or Γ



1, 2, 3 are from the DiVincenzo criteria for quantum computation.

Nobel Prize in Physics 2025



Ill. Niklas Elmehed ©
Nobel Prize Outreach
John Clarke

Prize share: 1/3



Ill. Niklas Elmehed ©
Nobel Prize Outreach
Michel H. Devoret

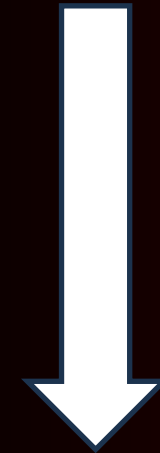
Prize share: 1/3



Ill. Niklas Elmehed ©
Nobel Prize Outreach
John M. Martinis

Prize share: 1/3

“for the discovery of
macroscopic quantum
mechanical tunnelling
and energy quantization
in an electric circuit”

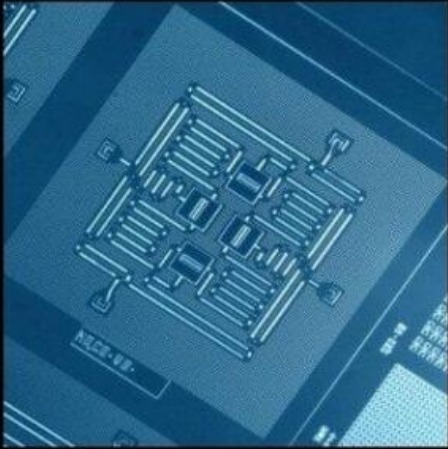


Single Cooper pair box
with two quantized
energy levels (Nakamura,
Pashkin, Tsai, 1999)

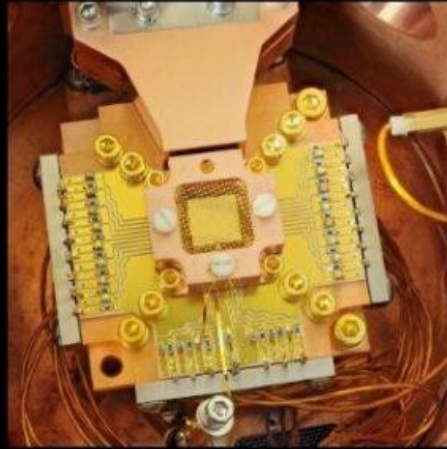
Superconducting
“transmon” qubits

<https://www.nobelprize.org/prizes/physics/2025/summary/>

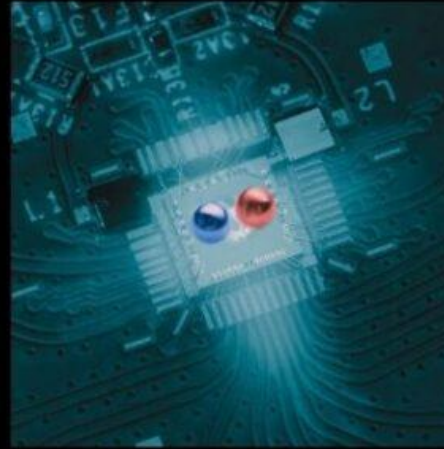
Types of qubits



SUPERCONDUCTING



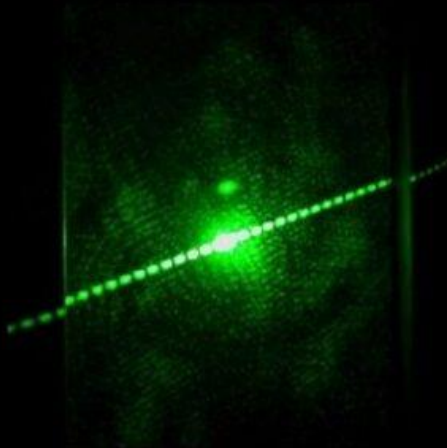
TRAPPED ION



SPIN



TOPOLOGICAL



PHOTONIC



NEUTRAL ATOM

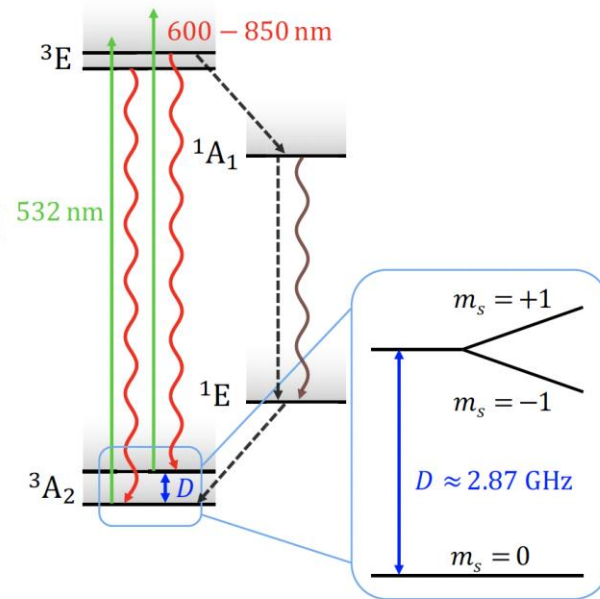
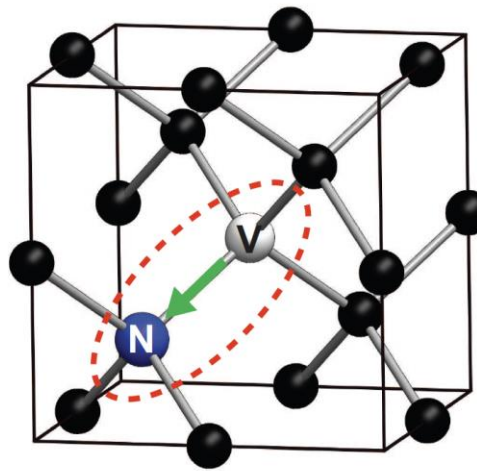
We don't ask,
“which qubit is
the best
computer?”
We ask, “which
qubit couples
most strongly to
the signal we are
looking for?”

Spin qubit example at room temperature

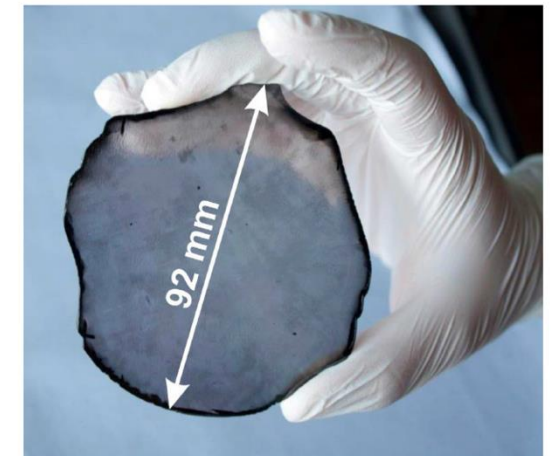
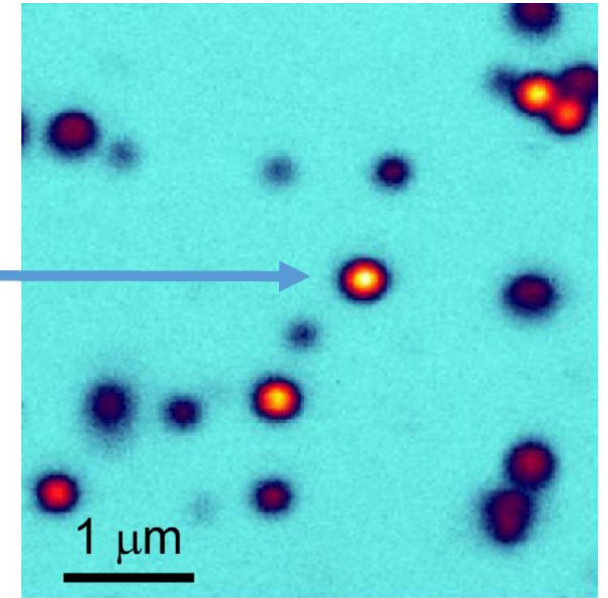
Nitrogen-Vacancy centers in diamond

[For reviews, see Doherty et al., 1302.3288; Barry et al. 1903.08176]

- In the ground state, e pair forms **spin triplet** at NV center
 - Control/measure spin states
- **Quantum sensing**
- Sensor as temperature, pressure, **magnetic field**, ...



Many applications : quantum physics, engineering, bioscience,... **particle physics!**



Hetero-epi, CVD, single crystal

Scientific reports 7:44462 (2017)
doi:10.1038/srep44462

Already useful without entanglement

NV Center in Diamond: Sensitivity and Tradeoff

Sensitivity (η)

$$\eta \propto \frac{1}{\sqrt{n_{NV} T_2}}$$

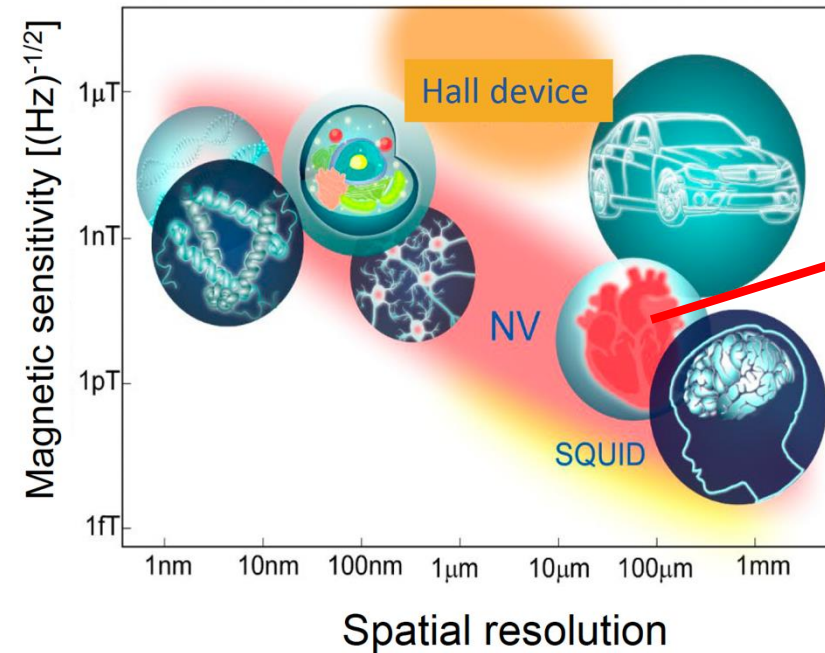
n_{NV} : Number of NV

T_2 : Relaxation time

O(1) ms achieved

Not only T_2 but also the number of NV contributes to the sensitivity.

- Caveat: higher concentration of NV centers results in shorter T_2 .
- Bigger sensor with a large number of NV centers for high sensitivity, keeping concentration manageable



J. F. Barry, D. A. Braje, et al.,
Phys. Rev. Appl. 22, 044069
(2024).

Mizuochi, OYO BUTSURI, 87, 251-261(2018).

“Light Dark Matter Search with Nitrogen-Vacancy Centers in Diamonds”

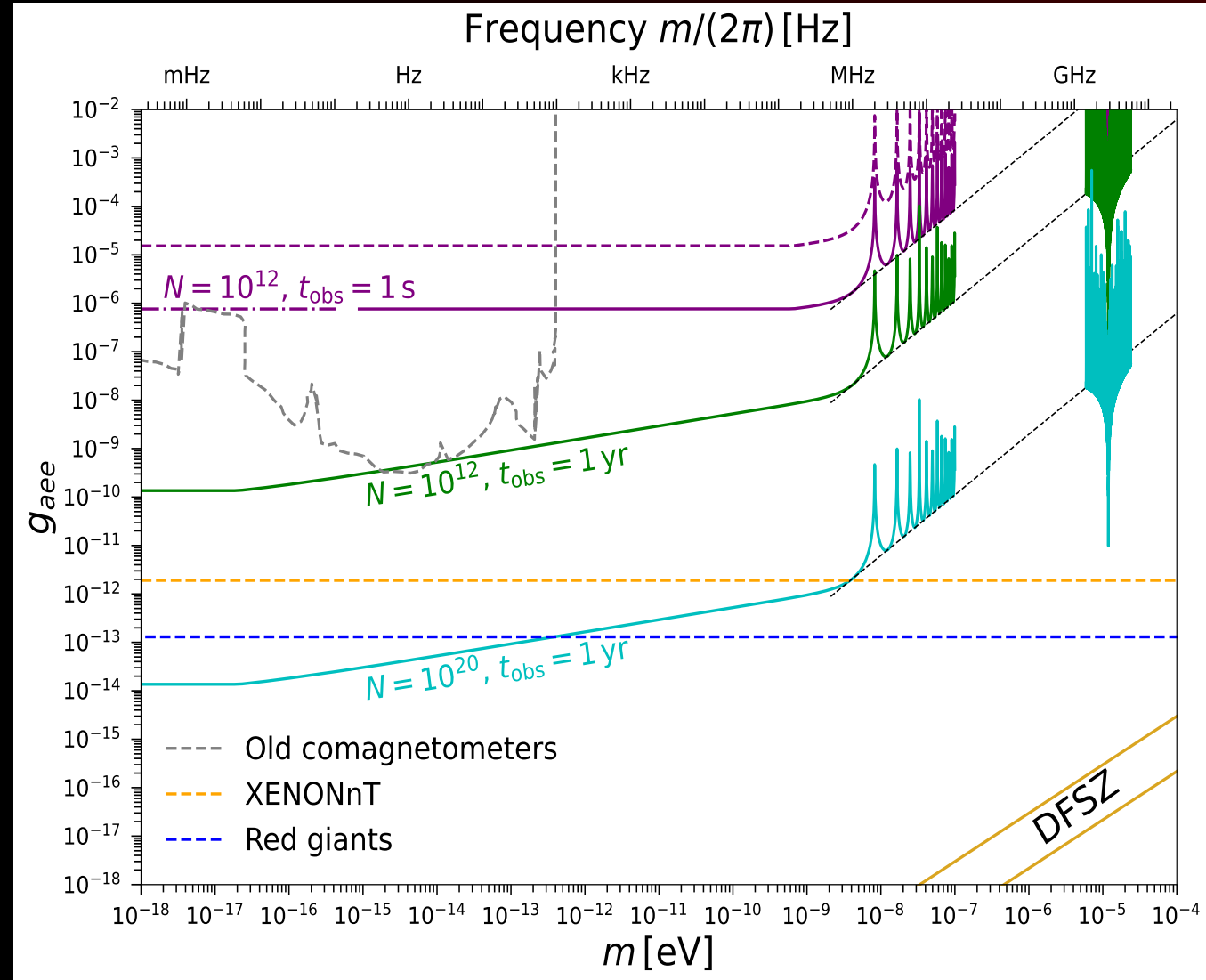
- Light Axion-Like Particles (ALPs) behave like a classical wave.

$$\phi(t) \simeq \phi_0 \cos(mt + m\vec{v} \cdot \vec{x}), \quad \phi_0 = \sqrt{2\rho_{\text{DM}}/m^2}$$

- ALP fields are similar to magnetic fields.

$$H_1 = \frac{g_e}{m_e} \vec{\nabla} \phi \cdot \vec{\sigma} \simeq \frac{g_e \sqrt{2\rho_{\text{DM}}}}{m_e} \cos(mt) \vec{v} \cdot \vec{\sigma}$$

- Broad mass reach! (Ramsey sequence limited by spin projection noise)
- Many experimental challenges remain, but no show stopper.
- Future measurements with scalable NV diamond quantum sensors are promising.



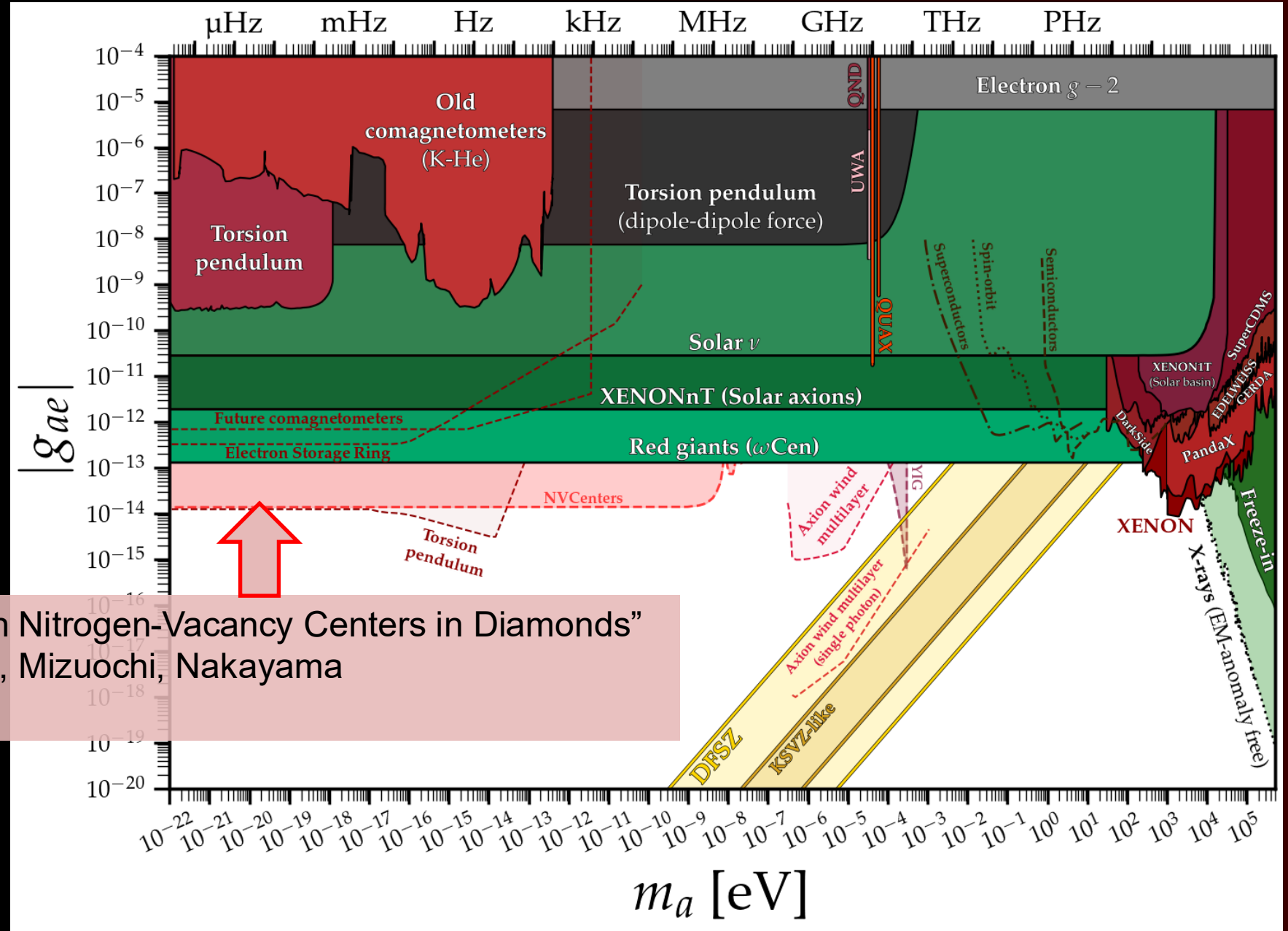
Chigusa, Hazumi, Herbschleb, Mizuochi, Nakayama, JHEP 03 (2025) 083

Axion-electron coupling

Results and future prospects

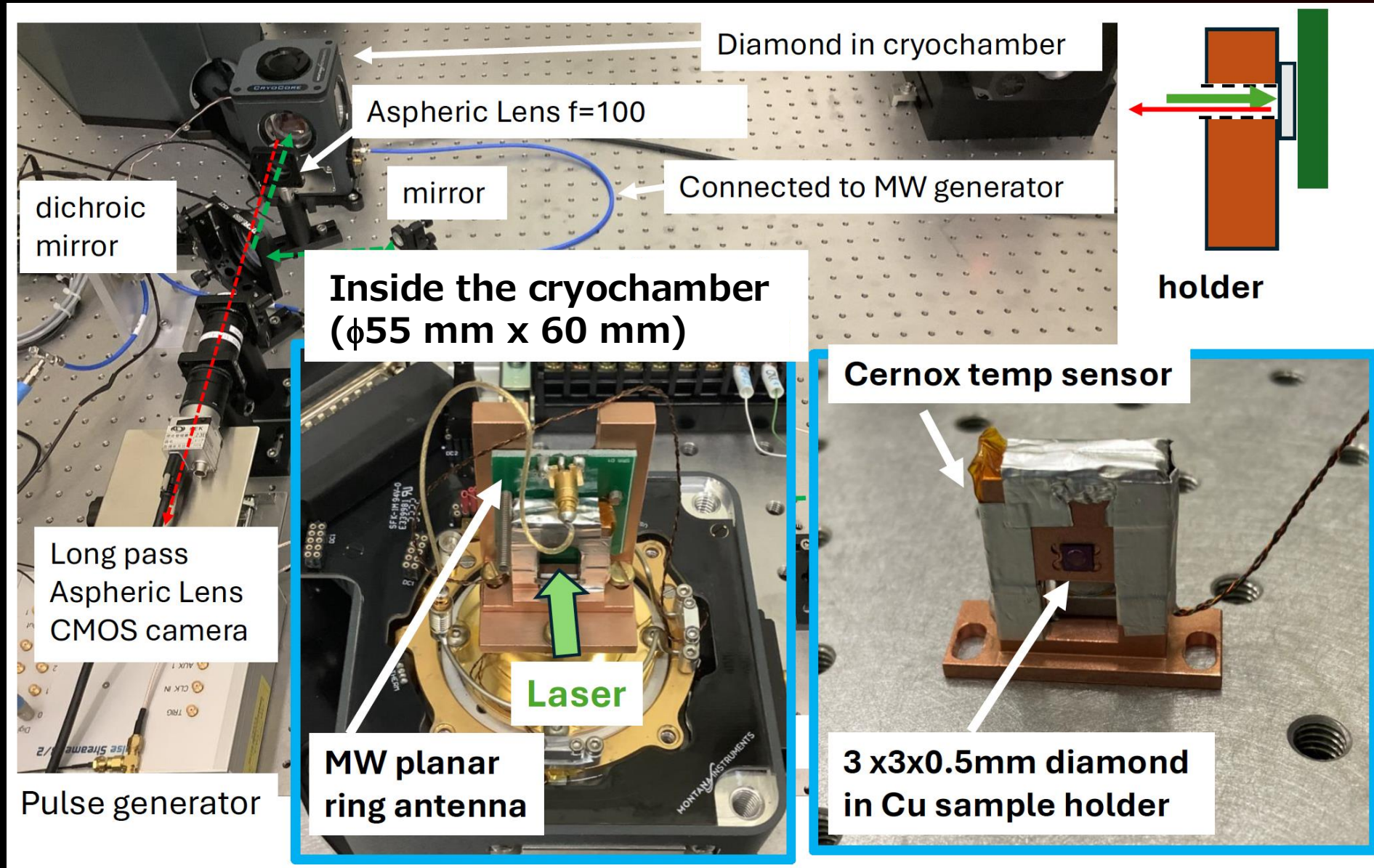
AxionLimits, compiled by C.O'Hare

“Light Dark Matter Search with Nitrogen-Vacancy Centers in Diamonds”
Chigusa, Hazumi, Herbschleb, Mizuochi, Nakayama
JHEP 03 (2025) 083



<https://cajohare.github.io/AxionLimits/>

Cryogenic system in preparation at KEK



Atsuhiro Umemoto

Aiming for the world's longest relaxation time T_2 as the first step

Nuclear Spin Metrology with Nitrogen Vacancy Center in Diamond for Axion Dark Matter Detection

Chigusa, Hazumi, Herbschleb, Matsuzaki, Mizuochi, Nakayama, *Phys.Rev.D* 111 (2025) 7, 075028

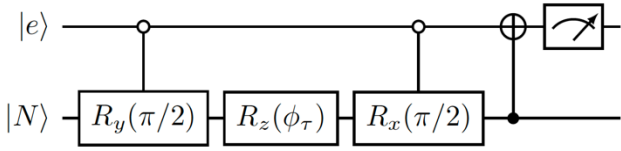
- ALP fields are similar to magnetic fields, **but different.**
No mass suppression!
- Nuclear spin metrology was “abandoned” in the QIS community, but very interesting for new physics search!
- Use of ^{14}N studied $\rightarrow$ next page

	axion	magnetic field
electron spin	$\mathcal{L} = \frac{1}{f} \vec{\nabla}_a \cdot \vec{S}_e$	$H \sim \frac{eg_e}{2m_e} \vec{B} \cdot \vec{S}_e$
nuclear spin	$\mathcal{L} = \frac{1}{f} \vec{\nabla}_a \cdot \vec{S}_N$	$H \sim \frac{eg_N}{2m_N} \vec{B} \cdot \vec{S}_N$
	Nuclear spin coupling is NOT suppressed	Nuclear spin coupling is suppressed

Nuclear Spin Metrology with Nitrogen Vacancy Center in Diamond for Axion Dark Matter Detection

Chigusa, Hazumi, Herbschleb, Matsuzaki, Mizuochi, Nakayama, *Phys.Rev.D* 111 (2025) 7, 075028

DC magnetometry (Ramsey sequence)

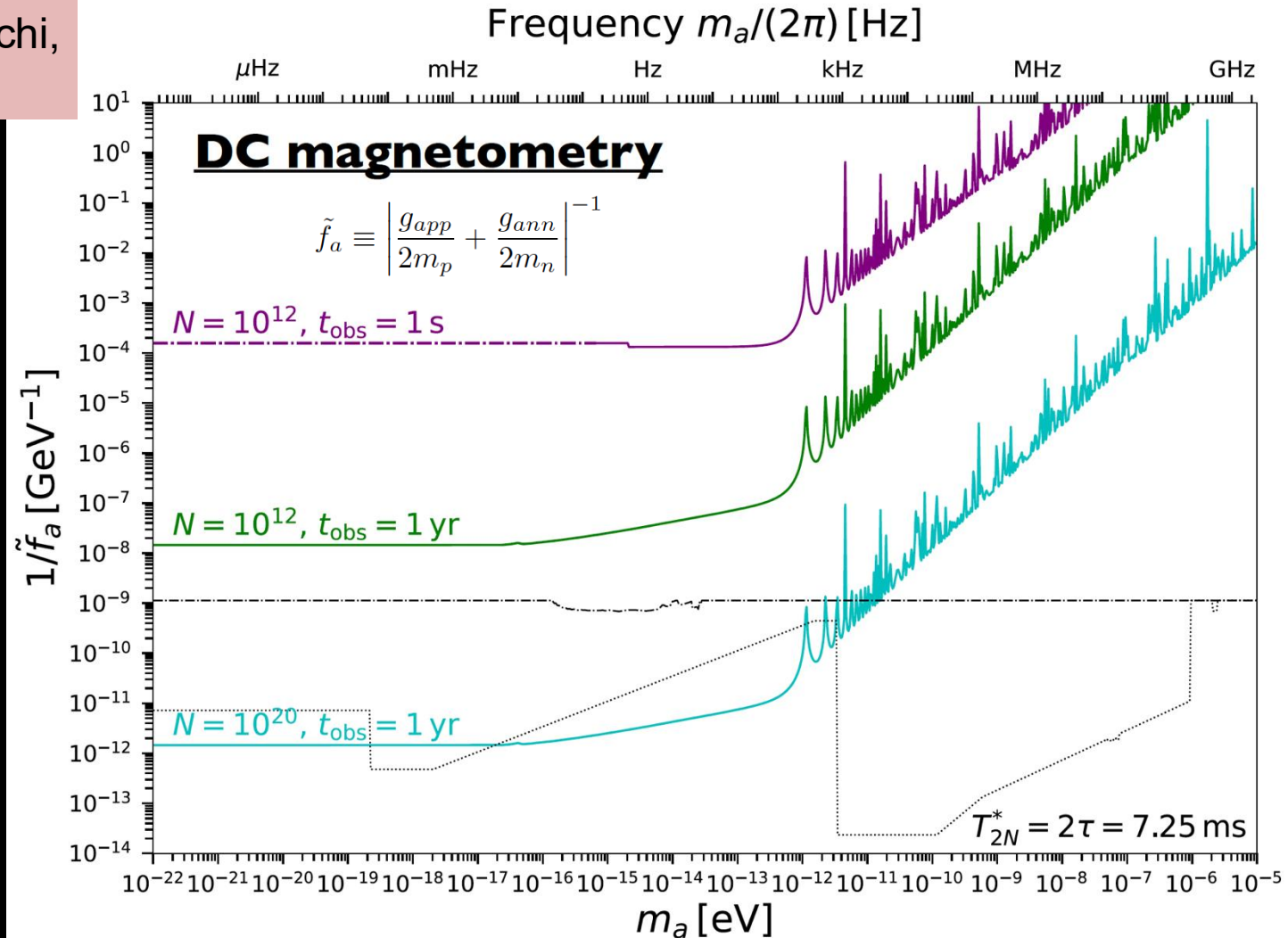


- Effective B field proportional to

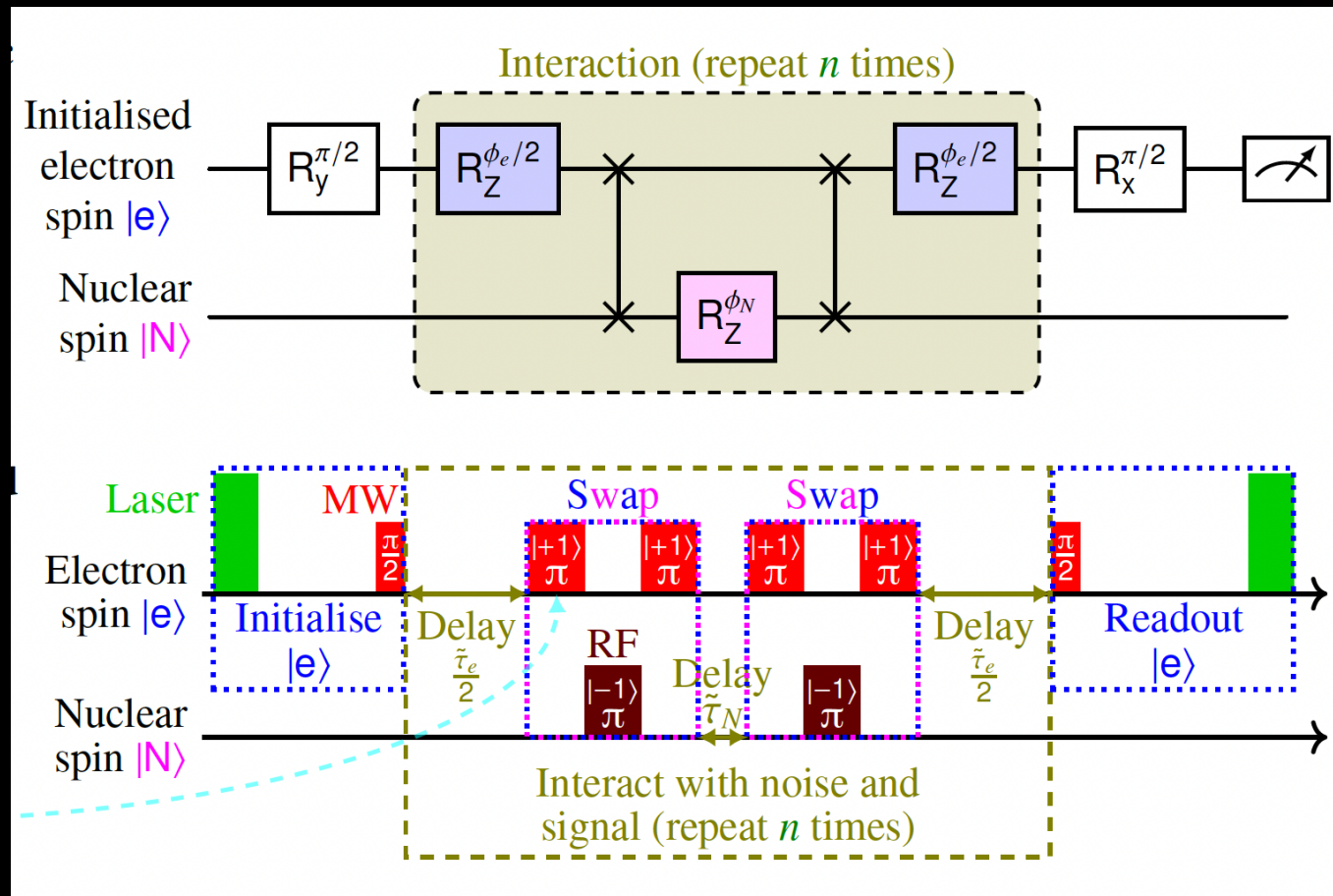
$$\frac{g_{app}}{m_p} + \frac{g_{ann}}{m_n} \quad \text{for } ^{14}\text{N}$$

$$\frac{g_{app}}{m_p} \quad \text{for } ^{15}\text{N}$$

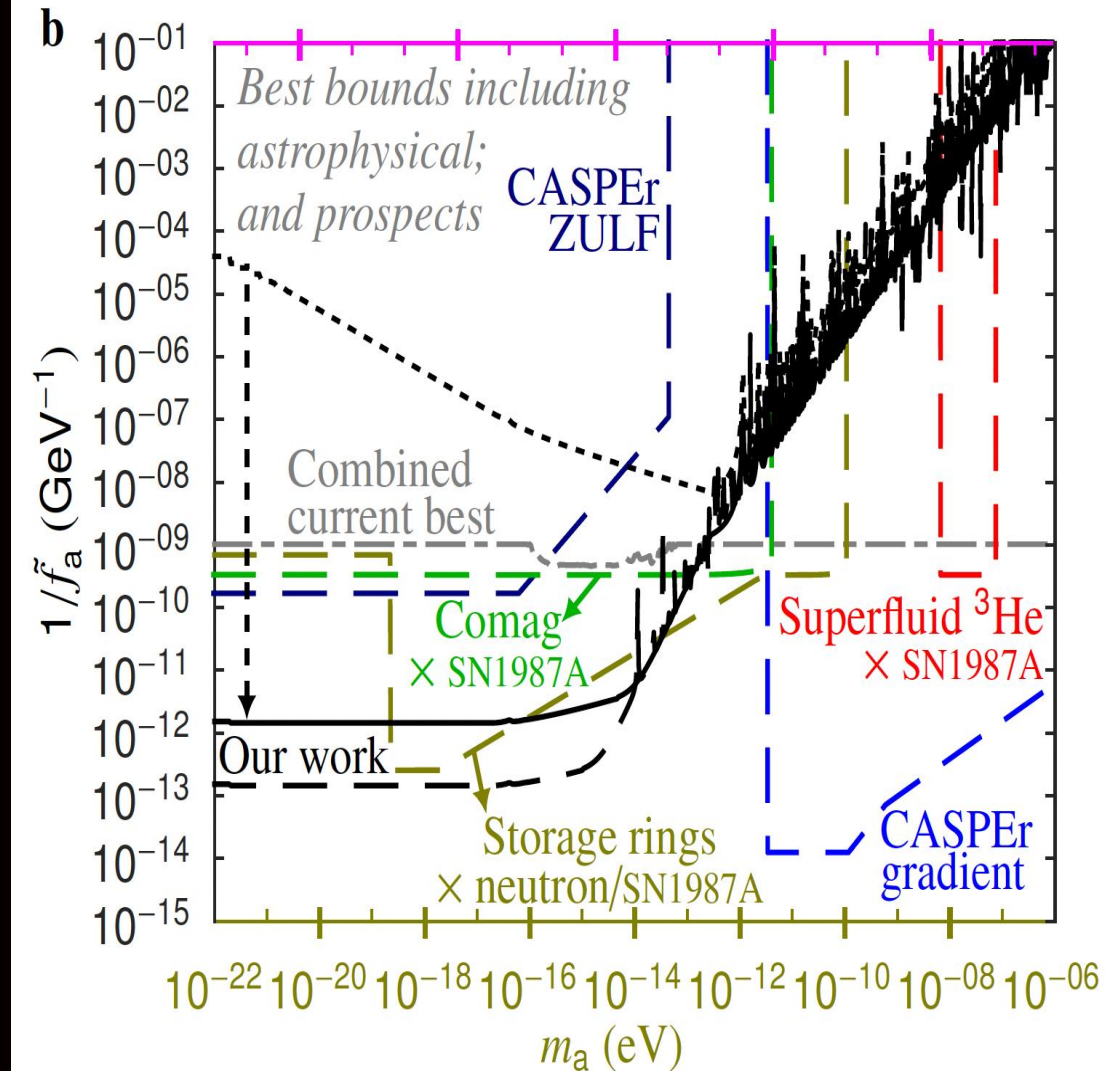
- ^{15}N -rich samples give constraints on g_{app} alone.
- Broad mass coverage!



Improved quantum protocol



"Hybrid-spin decoupling for noise-resilient DC quantum sensing,"
 S. Chigusa, MH, E. D. Herbschleb, Y. Matsuzaki, N. Mizuochi,
 K. Nakayama, arXiv:2511.16732



The most exciting quantum resource
for sensing is **quantum entanglement**.

Can we use it to search for dark matter?

N entangled qubits as quantum sensors: a conceptual example with spin qubits

Wineland-Bollinger-Itano-Heinzen 1994
Huelga et al. 1997

Initialization to the
Greenberger-Horne-Zelinger (GHZ) state

$$|\Psi(t=0)\rangle \equiv |\Psi_0\rangle = \frac{1}{\sqrt{2}} |00 \dots 0\rangle + \frac{1}{\sqrt{2}} |11 \dots 1\rangle.$$

Spins in a magnetic field

$$H \equiv \sum_{j=1}^N \frac{\omega}{2} \hat{\sigma}_z^{(j)}.$$

Lamor precession

$$\begin{aligned} |\Psi(t)\rangle &= e^{-iHt} |\Psi_0\rangle \\ &= \frac{1}{\sqrt{2}} e^{i\frac{\omega}{2}Nt} |00 \dots 0\rangle + \frac{1}{\sqrt{2}} e^{-i\frac{\omega}{2}Nt} |11 \dots 1\rangle \end{aligned}$$

Fast phase rotation!

Measurement by projection

$$\hat{P}_\perp \equiv |\Psi_\perp\rangle \langle \Psi_\perp|, \quad |\Psi_\perp\rangle \equiv \frac{1}{\sqrt{2}} |00 \dots 0\rangle - i \frac{1}{\sqrt{2}} |11 \dots 1\rangle$$

Measurement error of the
magnetic field B (proportional to ω)
is **inversely proportional to N**.
(ignoring decoherence)

$$|\langle \Psi(t) | \hat{P}_\perp | \Psi(t) \rangle|^2 = \frac{1}{2} + \frac{1}{2} \sin N\omega t \simeq \frac{1}{2} + \frac{1}{2} N\omega t,$$

In case of N **separable spins**,
the precision is **inversely proportional to N**.

Great sensitivity enhancement,
 $N/\sqrt{N} = \sqrt{N}$, with entanglement!

Discussion

- Issue

- Relaxation time of entanglement $\tau_{\text{GHZ}} \sim T_2/N$
→ lose merit of N -entangled state over separable N spins

- Merit

- Mass range of search: $\Delta m \sim 1/\tau_{\text{GHZ}} \sim N/T_2$
- Scan time can be reduced significantly

"Entanglement-enhanced AC magnetometry in the presence of Markovian noise," T. Sichanugrist, H. Fukuda, T. Moroi, K. Nakayama, S. Chigusa, N. Mizuochi, MH, Y. Matsuzaki, Phys.Rev.A 111 (2025) 4, 042605

- Comment

- Statistical inference for quantum sensing, or quantum inference, is a new area of research, also deeply connected to cosmology.
One possible way to keep Andrew busy throughout his new Zodiac cycle.

Summary

- Use of quantum sensors for dark matter search is a new research topic. Many new ideas are emerging!
- We can leverage rapid progress and large investment on Quantum 2.0.
- We hope to find some transformative applications for cosmology and particle physics. The exploration has just started!